



Brave

Jennifer Lopez's new album titled Brave will be sold on a chick-looking 1-GB flash drive

NAND mass production

Toshiba and SanDisk plan to mass produce NAND flash chips on 20nm-class process technology

How stuff works

NAND		
IN	IN	OUT
0	0	1
0	1	1
1	0	1
1	1	0

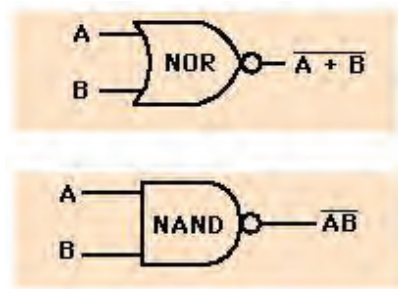
NOR		
IN	IN	OUT
0	0	1
0	1	0
1	0	0
1	1	0

NAND and NOR functions permit a variety of Boolean operations

are immensely useful for the creation of universal Boolean operations.

NAND – which is short for 'Not AND' refers to the Boolean logic operation that says that if any one input is false, the conclusion is true. Since computers only understand zeroes and ones, in the language of electronics, if any one input is a 0 (zero), the output must be a 1 (one). NOR (short for 'Not OR') is a Boolean which is true if none of the inputs are true – i.e. if any – or both – of the two inputs is a one (1), the result is a zero. Both of these can be used to make NOT, AND, OR XOR or XNOR, and so they're called 'Universal Gates' – and that's what makes them useful.

NAND and NOR gates are so designed as to function in the above manner. As you can see in the figure, the Control Gate in every



NOR and NAND gates

cell is connected to the horizontal 'Word Line' (WL). The 'Bit Line' (BL) connects all the drains together – which is actually the data bus of the chip. The Source Line (SL) connects all the sources to the ground.

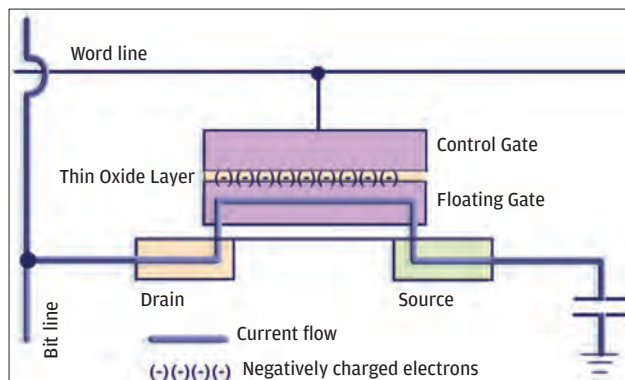
NAND vs NOR

NAND and NOR basically differ in architecture and they both have their pros and cons. NAND has its cells arranged in a series – like a NAND gate – and therefore lets you erase and write data faster, uses less space (about 60 per cent of a NOR flash) because it doesn't need so many 'grounds', cuts down on costs and is more robust than a NOR flash. NOR-flash memory, on the other hand, has every cell connected to the ground in parallel to the bit lines and another end to a bit line. This parallel connection is an advantage as every byte can be addressed individually. Yet, NOR is slower in writing and erasing but quicker than NAND in reading data. NAND flash requires that you read, erase or write data

on a block wise basis, where a collection of bytes makes a 'page' and a collection of pages makes a 'block'. Actions are performed in chunks and therefore NAND is unsuitable as a replacement for the ROM in your PC which needs random access at the byte level.

How flash drives work

Any data storage device must be able to perform the three basic actions of Write, Read and Erase. Whether it's NAND or NOR, reading, writing or erasing are all controlled entirely by the voltage combination applied to the WL and the BL. In the case of flash drives, it's all about setting the status of a cell to 'charged' or 'not charged' – or alternatively, sensing whether the floating gate in individual cells is set to 0 or 1. If a floating gate has electrons, it hinders the flow of electrons through the transistor (from the control gate to the silicon element), thereby allowing less than 50 per cent of the current to pass through – implying a zero (0). Similarly, if there is no negative charge in the Floating Gate impeding the flow of current, more than 50 per cent is allowed to pass through – indicating a one (1). This can be controlled by deliberately manipulating the voltage applied at the Source, Drain or Control gate to change the delta value. If the drive hasn't been used at all – or if it's blank – all the cells are set to '1' because the flow of electrons is unhindered. Thus the state of the flash cell is determined by whether or not the Floating Gate contains electrons or not. If it does it's programmed, if it doesn't, it's erased or blank. The changing voltage at every cell is measured by a flash Memory Sense Amplifiers which use a differential amplifier to measure the disparity between the voltage generated by a particular cell current (the flash Bit line voltage) and a reference voltage value. If the Bit Line voltage is higher, it implies that the 'Data Out' signal is pulled to a logic high (binary digit 1) and conversely, if it is lower, the 'Data Out' signal is pulled to a logic low (binary digit 0). Having understood this,



This one is set to zero

let's figure out how flash drives perform their individual tasks.

Erasing

Since none of the Floating Gates (FG) contain negative charge to begin with, all the bits will be set to '1'. Ergo, if you want to erase a programmed cell, you'll need to set it back to '1' – or rather, compel the FG to empty itself of electrons. Applying a voltage (between about 9V and 12V) from the source to the control gate along a Word Line would cause the electrons trapped in the Floating Gate to break out and 'tunnel' to the Source and the p-type substrate. The Floating Gate is now free of electrons and set again to '1'.

Programming

Given that all the cells are already at '1', programming a blank flash memory device becomes restricted to setting a cell to the binary '0' value. To write into a flash drive, then, a high voltage (about 7V) is applied to the Bit Line (the Drain) and a higher one (around 12V) to the Word Line (the Source). Electrons will now flow from the Source to the Drain. Typically, some over-excited electrons will jump and tumble into the FG (remember the 'Hot Carrier Injection' process?), getting trapped there, giving a negative charge. Turn the power off and the electrons still stay there, thanks to the insulation of the oxide around the FG. Thus, entire blocks of cells can be programmed into millions of sequences of zeroes and ones, literally, at the speed of light.

Reading

When a current is passed from the Control Gate (about 5V) to the Drain (about 1V), the current begins to flow between the two. The voltage is not high enough to bump electrons into the FG if it is not negatively charged but high enough to detect insulation or conduction at every cell. The cell sensor tracks the current flow and identifies whether a particular cell is conducting or remains insulating.

Accordingly, it arrives at a binary code of ones (1s) for every conducting cell and zeroes (0s) for every insulating one.

Types of flash drives

When it comes to flash storage devices these days, NOR flash takes a back seat and it seems to NAND all the way. Two basic forms of NAND are the preferred form of storage in solid state